Fintech Ecosystem Blueprint – Loan Wallet Settlement

Week-by-Week Implementation Checklist (Phase 1–6)

Phase 1 – Foundation (Weeks 1–4)

Goal: Menyiapkan pondasi arsitektur dan komponen dasar.

Week 1: - Setup Git repository + branch strategy (main, develop, feature/*) - Setup CI/CD skeleton (GitHub Actions or Jenkins pipeline) - Setup base Spring Boot multi-module project: - common (utils, base classes) - api-gateway - auth-service - customer-service - Integrate PostgreSQL & Flyway migration - Setup Dockerfile + docker-compose for local testing

Week 2: - Implement Authentication & Authorization (JWT + Spring Security) - Implement Role & Permission structure (Admin, Customer, Operator) - Setup base test framework (JUnit5 + Testcontainers) - Setup global exception handler & response wrapper

Week 3: - Implement Customer Registration & KYC mock flow - Add simple REST endpoint for onboarding (email verification, etc.) - Configure Swagger/OpenAPI docs per service

Week 4: - Integrate Service Registry (Eureka or Consul) - Add Config Server (Spring Cloud Config) - Deploy minimal setup to Kubernetes (RKE2 or Minikube) - Setup Helm chart for each microservice

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Phase 2 – Core Wallet System (Weeks 5–8)

Goal: Membangun sistem dompet digital dan transaksi dasar.

Week 5: - Create wallet-service (entity: Wallet, WalletTransaction) - Implement wallet creation on customer onboarding - Add credit/debit APIs + validation rules

Week 6: - Add idempotency layer using request ID + Redis cache - Implement Outbox pattern for event publishing - Add Kafka setup for event streaming (transaction events)

Week 7: - Integrate with auth-service for secured transaction endpoints - Add balance inquiry & transaction history APIs - Setup audit logging (via AOP or Kafka sink)

Week 8: - Load testing with 1K concurrent transaction simulation - Tune PostgreSQL (connection pool, indexing) - Enhance CI/CD pipeline (test → build → deploy)

Phase 3 – Loan Origination (Weeks 9–12)

Goal: Membangun proses pinjaman dari pengajuan sampai pencairan.

Week 9: - Create loan-service (entities: LoanApplication, LoanOffer, LoanAgreement) - Implement loan application endpoint (customer submits form)

Week 10: - Implement scoring-service (mock rules engine) - Connect to external credit check API (dummy first) - Publish events for scoring result

Week 11: - Implement loan approval flow + document signing simulation - Add integration with wallet-service for fund disbursement

Week 12: - Integrate Kafka topics for loan lifecycle events - Add compensating transaction (Saga orchestration) - Conduct end-to-end flow test (apply → approve → disburse)

Phase 4 – Repayment & Billing (Weeks 13–16)

Goal: Menangani tagihan dan pelunasan pinjaman.

Week 13: - Create billing-service (entities: Bill, BillItem, PaymentSchedule) - Generate bills automatically from loan agreements

Week 14: - Implement repayment flow (auto-debit wallet → update bill status) - Add late fee & penalty calculation logic - Publish payment events via Kafka

Week 15: - Add reconciliation-service for cross-checking wallet vs billing records - Setup async job for daily reconciliation - Add retry & compensation policy (dead-letter topic)

Week 16: - Add dashboard for operator/admin to view repayment analytics - Conduct integration & UAT testing

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Phase 5 – Settlement & Accounting (Weeks 17–20)

Goal: Menangani pencatatan akuntansi, settlement, dan laporan keuangan.

Week 17: - Create accounting-service (entities: JournalEntry, Ledger, Account) - Setup double-entry bookkeeping model (debit/credit)

Week 18: - Integrate accounting-service with wallet & billing-service via event sourcing - Add settlement job for end-of-day ledger balancing

Week 19: - Implement financial reporting (daily summary, GL report) - Optimize query with materialized views

Week 20: - Integrate Grafana + Prometheus for monitoring financial metrics - Conduct reliability test (chaos monkey simulation)

Phase 6 – Risk & Compliance (Weeks 21–24)

Goal: Memastikan sistem sesuai regulasi, aman, dan siap audit.

Week 21: - Add audit-service for full event logging (data lineage) - Implement transaction traceability (correlation ID across services)

Week 22: - Add compliance-service (AML/KYC screening simulation) - Integrate sanction list & transaction anomaly detection rules

Week 23: - Conduct penetration & security testing - Add data encryption at rest (PostgreSQL + Vault) - Add API rate limiting & request signing (HMAC)

Week 24: - Perform external audit dry-run (ISO/PCI DSS simulation) - Write full architecture & compliance documentation - Final presentation + demo deployment to cloud (AWS or GCP)

✅Deliverables per Phase

Each phase should produce: - Running microservices with test coverage ≥80% - Helm charts + CI/CD pipeline - Swagger documentation - Technical README + architecture diagram - Postman collection + demo video

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Next suggestion: create **learning companion guide** (skill focus, tools mastery, and checklist per week). Mau gw tambahkan bagian itu selanjutnya?